

Selected Crypto Assets: Bitcoin (BTC) and Ethereum (ETH)

This study focuses on Bitcoin (BTC) and Ethereum (ETH) as the primary crypto assets for quantitative trading. These two assets jointly account for the majority of crypto market capitalization, trading volume, and derivatives open interest, and they form the core risk drivers of the broader crypto ecosystem. More importantly, BTC and ETH exhibit distinct statistical and structural characteristics, making them suitable benchmarks for systematic trading rather than interchangeable proxies .

Statistical Features

1. Return Distribution and Volatility Characteristics

BTC and ETH returns exhibit strong deviations from normality, characterized by volatility clustering, heavy tails, and excess kurtosis. Compared with equity indices such as the S&P 500, Bitcoin returns show a significantly higher probability of extreme outcomes, confirming the presence of fat-tailed risk documented in the lecture material . Volatility is regime-dependent: during stress periods, annualized volatility frequently exceeds levels observed in traditional asset classes, while calmer regimes exhibit relative compression.

ETH generally displays higher realized volatility and beta than BTC, reflecting its closer linkage to protocol-level events (e.g., upgrades, staking flows, DeFi activity). This statistical asymmetry suggests that BTC is more suitable for macro-driven or volatility strategies, while ETH may better support higher-frequency or event-sensitive approaches.

2. Correlation Structure and Diversification Limits

Crypto markets exhibit structurally high cross-asset correlation, especially during drawdowns. Empirical correlation maps in the lecture slides show that BTC, ETH, and major altcoins often move in lockstep, with correlations exceeding 0.7–0.8 during stress regimes . As a result, traditional diversification across crypto assets frequently fails precisely when risk management is most needed. For quantitative trading, this implies that portfolio construction must rely on factor exposure control or hedging, rather than naïve asset-level diversification.

3. Market Microstructure and Trading Hours

Unlike equities, BTC and ETH trade in a continuous 24/7 global market. Empirical evidence shows that returns and volatility vary across U.S., European, and Asian trading hours, reflecting differences in liquidity provision, information flow, and participant composition . These persistent intraday patterns suggest the presence of time-zone-specific effects that can be exploited through segmented execution or intraday statistical models.

Risk Factors and Quantitative Implications

1. Tail Risk and Liquidation Dynamics

BTC and ETH are subject to severe downside tail risk, largely driven by leverage in perpetual futures markets. Forced liquidations can trigger self-reinforcing price cascades, producing sharp, discontinuous moves. From a quantitative perspective, this necessitates volatility-scaled position sizing, strict drawdown controls, and regime-aware risk filters, rather than static exposure.

2. Liquidity and Leverage Risk

Although BTC and ETH are the most liquid crypto assets, liquidity can deteriorate rapidly during stress events. High leverage availability amplifies short-term volatility and invalidates assumptions of continuous execution. Quant strategies must therefore incorporate liquidity-adjusted execution models and avoid overfitting to calm-market conditions .

3. Structural and Regulatory Risk

BTC and ETH trade across a fragmented ecosystem of spot exchanges, futures markets, options venues, and ETFs. Structural frictions include:

- a. ETF tracking error relative to spot BTC
- b. Futures contango/backwardation, affecting carry and roll yield
- c. Cross-exchange price dislocations, especially during volatility spikes

These frictions introduce additional sources of basis risk but also create opportunities for relative-value and arbitrage-oriented strategies.

4. Derivatives Market and Forward-Looking Signals

BTC and ETH possess the deepest derivatives markets in crypto, including CME futures and liquid options markets. Option-implied volatility frequently exhibits smile or smirk patterns, indicating that market participants assign elevated probabilities to extreme price movements . Funding rates, term structure, and implied volatility surfaces provide forward-looking information that can be incorporated into systematic trading signals and regime classification.

Why BTC and ETH Are Suitable for Quantitative Trading

BTC and ETH are attractive quantitative trading instruments because:

- a. Their volatility and correlation structures are persistent yet regime-dependent, enabling statistical modeling.
- b. Their derivatives markets generate observable risk premia and forward-looking signals.
- c. Their market structure repeatedly produces non-random, testable patterns, rather than purely idiosyncratic price moves.

At the same time, strong non-stationarity and tail risk require disciplined, rule-based strategies consistent with quantitative trading principles emphasized in the course.

Conclusion

BTC and ETH are not merely large or liquid crypto assets; they are distinct statistical objects with exploitable quantitative properties. Their fat-tailed return distributions, regime-dependent volatility, high correlation under stress, and mature derivatives markets make them suitable—but demanding—targets for systematic trading. Effective strategies must explicitly account for tail risk, leverage-driven dynamics, and structural frictions while leveraging the rich statistical signals unique to crypto markets.